

Neotech Faculty of Diploma Engineering



LAB MANUAL

OF

“COMPUTER NETWORKING”

(DI03000051)

3rd Semester

Department of Computer Engineering

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CERTIFICATE

This is to certify that Mr. / Ms with
enrollment no.....Has
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the subject (DI03000051)..... .. From the
department of.....
During the Academic year.....



Date: Faculty Signature:

COMPUTER ENGINEERING
DIPLOMA 3rd SEM SUBJECT: CN
SUBJECT CODE:(DI03000051)

List of Practical

Sr. No	Name of the Experiment				Page No	Date of submission	Sign
1.	Connect computer using given topology with wired media. Assume six devices are arranged, if in: a) bus topology b) ring topology c) star topology d) mesh topology Find out number of cables(links), ports needed in each device and total number of ports needed in entire network for each of above stated topology.						
2.	Study about OSI model network Layers						
3.	Prepare and Test Straight UTP Cable and Cross UTP Cable.						
4.	Study and Test various Network devices available at Department/Institute. (Repeater, Hub, Switch, Bridge, Router and Gateway).						
5.	Determine whether following IPv4 addresses are valid or invalid. If valid IPv4 address then find class, Network and Host ID of an IPv4 address. If invalid IPv4 address then write reason for the same. a) 1.4.5.5 b) 75.45.301.14 c) 111.56.045.78 d) 192.226.12.11 e) 130.45.151.154 f) 11100010.23.14.67 g) 221.34.7.8.20 h) 240.230.220.89						
6.	Determine Class and Network Address for given IPv4 address and subnet mask.						
	IPv4 address	Subnet Mask	Class	Subnet			

	172.16.2.10	255.255.255.0					
	10.6.24.20	255.25.5.240.0					
	10.30.36.12	255.255.255.0					
7.	Subnet the IP address 216.21.5.0 into 30 hosts in each subnet						
8.	Identify valid IPv6 addresses and if invalid IPv6 address then write reason for the same: a) 2001 : db8: 3333 : 4444 : 5555 : 6666 : 7777 : 8888 b) :: c) 225.1.4.2 d) 2001: db8: : e) :: 1234 : 5678 f) 2001 : db8: : 1234 : 5678 g) 2001:0db8:0001:0000:0000:0ab9:C0A8:0102 h) fe80:2030:31:24						
9.	Study of firewall in providing network security.						
10.	Runbasicutilitiesandnetworkcommands:ipconfig,ping, Tracert, netstat, pathping , route						
11.	List different adapters in any computer/ laptop and configure network connection in your institute/ premises LAN.						

Practical – 1: Connect computer using given topology with wired media. Assume six devices are arranged, if in: a) bus topology b) ring topology c) star topology d) mesh topology Find out number of cables(links), ports needed in each device and total number of ports needed in entire network for each of above stated topology.

Solution:

1. **Bus Topology:**

- In a bus topology, all devices are connected to a single central cable (the bus). There are no intermediate devices like switches or hubs.
- **Number of Cables (Links):** Since there's only one shared cable, we need **5 cables** (connecting 6 devices minus 1).
- **Ports Needed in Each Device:** Each device requires **2 ports**: one for connecting to the bus and another for the internal network interface.
- **Total Ports Needed in Entire Network:** 6 devices $\times$ 2 ports/device = **12 ports**.

2. **Ring Topology:**

- In a ring topology, devices are connected in a circular manner, forming a closed loop.
- **Number of Cables (Links):** We need **6 cables** (one between each adjacent pair of devices).
- **Ports Needed in Each Device:** Each device requires **2 ports**: one for connecting to the left neighbour and another for the right neighbour.
- **Total Ports Needed in Entire Network:** 6 devices $\times$ 2 ports/device = **12 ports**.

3. **Star Topology:**

- In a star topology, all devices are connected to a central hub or switch.
- **Number of Cables (Links):** We need **6 cables** (one from each device to the central hub/switch).
- **Ports Needed in Each Device:** Each device requires **1 port** to connect to the central hub/switch.
- **Total Ports Needed in Entire Network:** 6 devices $\times$ 1 port/device = **6 ports**.

4. **Mesh Topology:**

- In a full mesh topology, every device is directly connected to every other device.
- **Number of Cables (Links):** Since there are 6 devices, we need to calculate the total number of connections (links) between them.
 - The formula for the total number of connections in a full mesh is: $n(n-1)/2$, where **n** is the number of devices.
 - For 6 devices: $6(6-1)/2 = 15$ connections.
 - So, we need **15 cables**.
- **Ports Needed in Each Device:** Each device requires **5 ports** (one for each of the other 5 devices).
- **Total Ports Needed in Entire Network:** 6 devices $\times$ 5 ports/device = **30 ports**.

Practical – 2: Study about OSI model network Layers.

Solution:

The **OSI (Open Systems Interconnection) model**, which provides a structured framework for understanding how network communication occurs. The OSI model consists of **seven layers**, each with specific functions. Here's an overview of these layers:

1. **Physical Layer (Layer 1):**
 - The **lowest layer** in the OSI model.
 - Responsible for the **actual physical connection** between devices.
 - Deals with **bits** and their transmission.
 - Key functions:
 - **Bit synchronization:** Provides clock synchronization for bits.
 - **Bit rate control:** Defines the transmission rate (bits per second).
 - Specifies **physical topologies** (e.g., bus, star, mesh).
 - Determines **transmission modes** (simplex, half-duplex, full-duplex).
 - [Devices: Hub, Repeater, Modem, and Cables](#)
2. **Data Link Layer (Layer 2):**
 - Ensures **error-free data transfer** between nodes.
 - Transmits packets from one node to another over the physical layer.
 - Divided into two sublayers:
 - **Logical Link Control (LLC):** Manages communication between devices.
 - **Media Access Control (MAC):** Handles addressing and framing.
 - [Devices: Switches, NICs \(Network Interface Cards\)](#)
3. **Network Layer (Layer 3):**
 - Responsible for **routing** and **logical addressing** (IP addresses).
 - Determines the best path for data packets to reach their destination.
 - Key functions:
 - **Routing:** Selects optimal routes.
 - **Logical addressing:** Assigns unique IP addresses.
 - [Devices: Routers, Layer 3 switches](#) .
4. **Transport Layer (Layer 4):**
 - Ensures **end-to-end communication** between devices.
 - Segments data into smaller units (segments or datagrams).
 - Provides **error detection, flow control, and reliability**.
 - [Protocols: TCP \(Transmission Control Protocol\), UDP \(User Datagram Protocol\)](#) .

5. **Session Layer (Layer 5):**

- Manages **sessions** (connections) between devices.
- Establishes, maintains, and terminates communication sessions.
- Handles **dialog control** (half-duplex, full-duplex).
- [Devices: Not directly associated with specific hardware](#)

6. **Presentation Layer (Layer 6):**

- Focuses on **data representation** and **encryption**.
- Translates data formats between sender and receiver.
- Handles **encryption, compression, and data conversion**.
- [Devices: Not directly associated with specific hardware](#) .

7. **Application Layer (Layer 7):**

- The **topmost layer** that interacts directly with end-users and applications.
- Provides **network services** to applications.
- Examples: HTTP, FTP, SMTP, DNS.
- [Devices: Not directly associated with specific hardware](#) .

Practical – 3: Prepare and Test Straight UTP Cable and Cross UTP Cable.

Solution:

Certainly! Let's dive into creating and testing both **straight-through** and **crossover** Unshielded Twisted-Pair (UTP) Ethernet cables:

1. **Straight-Through Cable:**

- A straight-through cable is used to connect different types of devices, such as a computer to a switch or router.
- **Color Coding:**
 - Both ends of the cable follow the same color sequence (pin 1 to pin 8).
 - The standard color order is: **white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.**

○ **Steps to Prepare and Test a Straight-Through Cable:**

1. **Obtain and Prepare the Cable:**

- Get a UTP cable (Cat 5e or Cat 6).
- Strip the outer insulation to expose the individual wires.

2. **Insert the Wires:**

- Arrange the wires according to the color code.
- Insert them into an RJ-45 connector in the correct order.

3. **Inspect, Crimp, and Re-inspect the Connector:**

- Ensure the wires are fully inserted and aligned.
- Use a crimping tool to secure the connector.
- Inspect for any loose wires or misalignment.

4. **Terminate the Other Cable End:**

- Repeat the same process for the other end of the cable.

5. **Test the Cable with a Wiremap Tester:**

- Use a wiremap tester to verify that the wires are correctly connected.

6. **Optional: Test the Cable Using a Cable Meter or Cable Certification Meter:**

This step is optional but recommended for professional installations.

7. Connect a Computer Using the Cable:

- Plug one end of the straight-through cable into your computer's network port.
- Plug the other end into a switch, router, or hub.

**2. Crossover Cable:**

- A crossover cable is used to connect similar devices directly, such as two computers or two switches.
- **Color Coding:**
 - The color sequence is different on each end of the cable.
 - One end follows the standard color order, while the other end swaps the orange and green pairs.

○ Steps to Prepare and Test a Crossover Cable:**1. Obtain and Prepare the Cable:**

- Get a UTP cable (Cat 5e or Cat 6).
- Strip the outer insulation to expose the individual wires.

2. Insert the Wires:

- Arrange the wires according to the color code for one end (standard order).
- For the other end, swap the orange and green pairs.

3. Inspect, Crimp, and Re-inspect the Connector:

- Ensure the wires are fully inserted and aligned.
- Use a crimping tool to secure the connector.
- Inspect for any loose wires or misalignment.

4. Terminate the Other Cable End:

- Repeat the same process for the other end of the cable.

5. **Test the Cable with a Wiremap Tester:**
 - Use a wiremap tester to verify that the wires are correctly connected.

6. **Optional: Test the Cable Using a Cable Meter or Cable Certification Meter:**
 - This step is optional but recommended for professional installations.

7. **Connect Similar Devices Using the Cable:**
 - Plug one end of the crossover cable into the network port of one device.
 - Plug the other end into the network port of the other device.

Practical – 4: Study and Test various Network devices available at Department/Institute. (Repeater, Hub, Switch, Bridge, Router and Gateway).

Solution:

1. **Repeater:**

- **Function:** Extends the range of a network by amplifying and retransmitting signals.
- **Use Case:** Boosts signal strength in long cable runs.
- **Example:** Simple signal booster.

2. **Hub:**

- **Function:** Connects multiple devices in a single network segment.
- **Use Case:** Basic connectivity; all data is broadcast to all devices.
- **Example:** Legacy Ethernet hubs.

3. **Switch:**

- **Function:** Efficiently forwards data only to the intended recipient.
- **Use Case:** Local area networks (LANs) with multiple devices.
- **Example:** Ethernet switches.

4. **Bridge:**

- **Function:** Connects two LAN segments, filtering traffic based on MAC addresses.
- **Use Case:** Segments LANs and reduces collision domains.
- **Example:** Wireless bridges.

5. **Router:**

- **Function:** Routes data between different networks (e.g., LAN to WAN).
- **Use Case:** Interconnects networks; performs IP routing.
- **Example:** Home Wi-Fi routers.

6. **Gateway:**

- **Function:** Translates data between different protocols or network types.
- **Use Case:** Connects LANs to external networks (e.g., internet).
- **Example:** Firewall gateways.

Practical – 5: Determine whether following IPv4 address are valid or invalid. If valid IPv4 address then find class, Network and Host ID of an IPv4 address. If invalid IPv4 address then write reason for the same.

- a) 1.4.5.5 b) 75.45.301.14 c) 111.56.045.78 d) 192.226.12.11 e) 130.45.151.154
f) 11100010.23.14.67 g) 221.34.7.8.20 h) 240.230.220.89

Solution:

1. **1.4.5.5:**

- **Validity:** Invalid (only one digit in each segment).
- **Reason:** IPv4 addresses must have four segments separated by dots.

2. **75.45.301.14:**

- **Validity:** Invalid (segment value exceeds 255).
- **Reason:** Each segment should be between 0 and 255.

3. **111.56.045.78:**

- **Validity:** Invalid (leading zeros in segment).
- **Reason:** Leading zeros are not allowed in segments.

4. **192.226.12.11:**

- **Validity:** Valid.
- **Class:** Class C (192.0.0.0 to 223.255.255.255).
- **Network ID:** 192.226.12.0.
- **Host ID:** 0.0.0.11.

5. **130.45.151.154:**

- **Validity:** Valid.
- **Class:** Class B (128.0.0.0 to 191.255.255.255).
- **Network ID:** 130.45.0.0.
- **Host ID:** 0.0.151.154.

6. **11100010.23.14.67:**

- **Validity:** Invalid (binary format).
- **Reason:** IPv4 addresses should be in decimal format

7. **221.34.7.8.20:**

- **Validity:** Invalid (extra segment).
- **Reason:** IPv4 addresses must have exactly four segments.

8. **240.230.220.89:**

- **Validity:** Valid.
- **Class:** Class E (reserved for experimental use).
- **Network ID:** 240.0.0.0.
- **Host ID:** 0.0.220.89.

Practical – 6: Determine Class and Network Address for given IPv4 address and subnet mask.

IPv4 address	Subnet Mask	Class	Subnet
172.16.2.10	255.255.255.0		
10.6.24.20	255.255.240.0		
10.30.36.12	255.255.255.0		

Solution:

1. **172.16.2.10:**

- **Class:** Class B (128.0.0.0 to 191.255.255.255).
- **Network ID:** 172.16.0.0.
- **Host ID:** 0.0.2.10.
- **Subnet Mask:** 255.255.255.0.

2. **10.6.24.20:**

- **Class:** Class A (0.0.0.0 to 127.255.255.255).
- **Network ID:** 10.0.0.0.
- **Host ID:** 6.24.20.
- **Subnet Mask:** 255.255.240.0.

3. **10.30.36.12:**

- **Class:** Class A (0.0.0.0 to 127.255.255.255).
- **Network ID:** 10.0.0.0.
- **Host ID:** 30.36.12.
- **Subnet Mask:** 255.255.255.0.

Practical – 7: Subnet the IP address 216.21.5.0 into 30 hosts in each subnet.

Solution:

Certainly! Let's subnet the IP address **216.21.5.0** into subnets with **30 hosts** in each subnet.

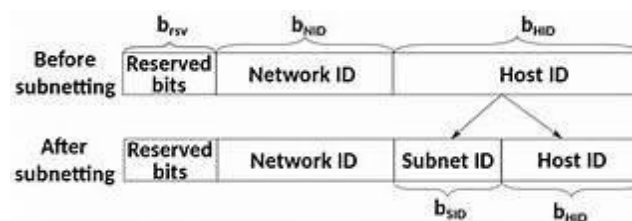
1. **IP Address:** 216.21.5.0
2. **Subnet Mask:** We need to determine the appropriate subnet mask to accommodate 30 hosts per subnet.

To achieve 30 hosts, we need at least 5 bits for host addresses (since $2^5 = 32$, which is more than 30). The remaining bits will be used for the network portion.

- **Network ID:** 216.21.5.0
- **Subnet Mask:** We'll use a /27 subnet mask (255.255.255.224), which provides 30 usable host addresses per subnet.

Now let's calculate the subnets:

1. **First Subnet:**
 - Network ID: 216.21.5.0
 - Broadcast Address: 216.21.5.31
 - Usable Host Range: 216.21.5.1 to 216.21.5.30
2. **Second Subnet:**
 - Network ID: 216.21.5.32
 - Broadcast Address: 216.21.5.63
 - Usable Host Range: 216.21.5.33 to 216.21.5.62
3. Continue this pattern for subsequent subnets.



Practical – 8: Identify valid IPv6 addresses and if invalid IPv6 address then write reason for the same:

- a) 2001 : db8: 3333 : 4444 : 5555 : 6666 : 7777 : 8888
- b) ::
- c) 225.1.4.2
- d) 2001: db8: :
- e) :: 1234 : 5678
- f) 2001 : db8: : 1234 : 5678
- g) 2001:0db8:0001:0000:0000:0ab9:COA8:0102
- h) fe80:2030:31:24

Solution:

Certainly! Let's evaluate each of the provided IPv6 addresses:

1. **a) 2001:db8:3333:4444:5555:6666:7777:8888:**
 - **Validity:** Valid.
 - **Reason:** This address follows the correct IPv6 format with eight segments, each containing hexadecimal digits.
2. **b) :::**
 - **Validity:** Valid.
 - **Reason:** The double colon (::) represents a series of consecutive zeros, which is allowed in IPv6 compression.
3. **c) 225.1.4.2:**
 - **Validity:** Invalid.
 - **Reason:** This address is in IPv4 format (decimal notation), not IPv6.
4. **d) 2001:db8:::**
 - **Validity:** Valid.
 - **Reason:** The address has the correct format with two segments and a double colon for zero compression.

5. e) ::1234:5678:
- **Validity:** Valid.
 - **Reason:** The address uses zero compression (::) and includes valid hexadecimal digits.
6. f) 2001:db8::1234:5678:
- **Validity:** Valid.
 - **Reason:** Properly compressed with zero compression (::) and valid hexadecimal digits.
7. g) 2001:0db8:0001:0000:0000:0ab9:C0A8:0102:
- **Validity:** Valid.
 - **Reason:** Correct format with eight segments, each containing valid hexadecimal digits.
8. h) fe80:2030:31:24:
- **Validity:** Valid.
 - **Reason:** Properly formatted IPv6 address with valid hexadecimal digits.

Practical – 9: Study of firewall in providing network security.

Solution:

Understanding **firewall technology** is essential for comprehending its role in safeguarding sensitive information and preventing unauthorized access. [The primary purpose of a firewall is to protect networks from unauthorized access, malicious activities, and dangerous cybersecurity threats¹.](#) Let's delve deeper into this critical network security component:

1. What Is a Firewall?

- A **firewall** acts as a barrier between secured internal networks and outside untrusted networks (such as the Internet).
- It monitors all incoming and outgoing traffic based on a defined set of security rules.
- Based on these rules, a firewall can:
 - **Accept:** Allow specific traffic.
 - **Reject:** Block traffic and reply with an “unreachable error.”
 - **Drop:** Block traffic without any reply.

2. How Does a Firewall Work?

- A firewall matches network traffic against its rule set.
- Once a rule is matched, the associated action (accept, reject, or drop) is applied.
- Rules can be defined based on the organization's security policies.
- Firewalls handle both **outgoing** and **incoming** traffic:
 - Outgoing traffic (originating from the server itself) is mostly allowed.
 - Incoming traffic is treated differently and analyzed based on transport layer protocols (TCP, UDP, or ICMP).
 - Default policies ensure consistent behavior when no specific rule matches.

3. Types of Firewalls:

- **Stateless Packet Filtering Firewalls:**
 - Examines individual packets without considering their context.
 - Decisions are based on source/destination IP addresses and port numbers.
- **Stateful Firewalls (Stateful Inspection Firewalls):**

Maintains a state table to track active connections

Considers the state of connections (e.g., established, related, new).

Provides better security by understanding the context of traffic.

4. Application Layer Firewalls (Proxy Firewalls):

- Operate at the application layer (Layer 7) of the OSI model.
- Inspect application-specific data (e.g., HTTP requests).
- Can provide content filtering and URL blocking.

5. Next-Generation Firewalls (NGFW):

- Combine traditional firewall features with intrusion prevention, application awareness, and more.
- Enhance security by analyzing traffic at deeper levels.

Practical – 10: Run basic utilities and network commands: ipconfig, ping, tracert, netstat, pathping , route.

Solution:

Here's a concise summary of the assignment related to basic networking utilities and commands for your 4th semester in diploma computer engineering:

1. **ipconfig:** This command displays the IP configuration details of your network interfaces on a Windows system. It provides information about IP addresses, subnet masks, default gateways, and more.
 2. **ping:** The ping command is used to test network connectivity by sending ICMP echo requests to a specified host. It measures the round-trip time and checks if the host is reachable.
 3. **tracert:** Also known as traceroute, this command traces the route taken by packets from your computer to a destination host. It shows the intermediate routers and their response times.
 4. **netstat:** The netstat command provides information about network connections, routing tables, interface statistics, and more. It's useful for diagnosing network issues.
 5. **pathping:** Similar to tracert, pathping combines traceroute and ping. It provides detailed information about network latency and packet loss along the route.
- router:** A router is a networking device that forwards data packets between different networks. It plays a crucial role in connecting devices within a network and to the internet.

Practical – 11: List different adapters in any computer/ laptop and configure network connection in your institute/ premises LAN

Solution:

1. Different Adapters in a Computer / Laptop

A **network adapter** is a hardware or software interface that connects a computer to a network. In a laptop or desktop, you'll usually find these types:

Common Network Adapters

1. **Ethernet Adapter (LAN Card / NIC)**
 - Wired adapter using an RJ-45 port.
 - Supports speeds like 100 Mbps / 1 Gbps / 10 Gbps.
2. **Wireless Adapter (Wi-Fi Card)**
 - Built-in or USB adapter.
 - Connects to WLAN using IEEE 802.11 standards (a/b/g/n/ac/ax).
3. **Bluetooth Adapter**
 - For personal area networking (PAN), file transfer, tethering, etc.
4. **USB Network Adapter**
 - External adapter plugged into a USB port.
 - Can be wired or wireless.
5. **Virtual Network Adapter (Software-based)**
 - Created by virtualization software (VMware, VirtualBox, Hyper-V).
 - Used for virtual machines or VPN connections.
6. **Cellular / Mobile Data Adapter (4G/5G Dongle or Built-in SIM slot)**
 - Provides internet via mobile networks.

2. Configuring Network Connection in LAN (Institute / Premises)

Step-by-Step LAN Configuration (Windows Example)

1. **Check Adapter**
 - Open *Control Panel* → *Network and Sharing Center* → *Change adapter settings*.
 - Identify *Ethernet* (wired) or *Wi-Fi* (wireless).
2. **Configure IP Address (Static or Dynamic)**
 - Right-click on adapter → *Properties*.
 - Select *Internet Protocol Version 4 (TCP/IPv4)* → *Properties*.
 - Choose:
 - **DHCP (Automatic IP):** Select *Obtain an IP address automatically*.
 - **Static IP (Manual):** Enter details like:
 - IP Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - Preferred DNS: 8.8.8.8

3. Connect LAN Cable / Wi-Fi

- Plug RJ-45 cable to switch/router for wired.
- Or connect to institute's Wi-Fi SSID.

4. Verify Connection

- Open *Command Prompt*.
- Run:
 - ipconfig → to check IP address.
 - ping 192.168.1.1 → to test gateway.
 - ping www.google.com → to test internet.